

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - Industry based Problem-Solving Question

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Industry based Problem-Solving Question	Assessment:	Assessment Tool 2– Industry based Problem-Solving Question
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	CSE-DS	Date:	20-08-2026

Code of Conduct

I Abzam A (192573029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

		reasoning and insights				
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

QUESTIONS:

16.CYBERSECURITY

Topic: Hypothesis Testing Using p-value

A cybersecurity system historically detects 95% of malicious activities. After deploying a new detection algorithm, it correctly detects 194 out of 200 attacks.

Test whether the new algorithm has significantly improved the detection rate using a p-value.

17.HEALTHCARE

Topic: Hypothesis Testing Using p-value

A hospital introduces a new appointment scheduling system. Before implementation, the average patient waiting time was 42 minutes. After implementation, a sample of 64 patients has an average waiting time of 38 minutes with a standard deviation of 12 minutes.

Test whether the new system significantly reduces waiting time using the p-value approach.

18. SOCIAL MEDIA ANALYTICS

Topic: Hypothesis Testing Using p-value

A social media platform claims that the average daily usage time of its users is 3 hours. A random sample of 100 users has an average usage time of 3.2 hours with a standard deviation of 0.8 hours.

Determine whether the data provide sufficient evidence to reject the company's claim at $\alpha = 0.05$.

19. MACHINE LEARNING

Topic: Hypothesis Testing

A data science team develops a new classification model. The existing model has an accuracy of 82%, while the new model achieves 86% accuracy on a randomly selected test sample of 500 observations.

Formulate suitable null and alternative hypotheses and determine whether the improvement is statistically significant using a p-value.

20. MANUFACTURING / QUALITY CONTROL

Topic: Confidence Interval (CI) + p-value

A manufacturer claims that its production process has a defect rate of only 2%. From 2,000 randomly selected products, 55 are defective.

Perform a hypothesis test at the 5% significance level using both:

(a) Confidence Interval approach

(b) p-value approach

Compare the conclusions.

NOTE: FOR EACH PROBLEM, FIND THE FOLLOWING

1. Identify the population and sample.
2. Identify the parameter of interest.
3. Calculate the point estimate.
4. State the null hypothesis (H_0).
5. State the alternative hypothesis (H_1).
6. Select the significance level (α).
7. Calculate the confidence interval and/or test statistic.
8. Calculate the p-value.
9. Compare the p-value with α .
10. Make an industry-oriented conclusion.

ANSWERS

16. Cybersecurity – Hypothesis Testing Using p-value

Problem: A cybersecurity system historically detects 95% of malicious activities. After deploying a new algorithm, it correctly detects 194 out of 200 attacks. Test whether the new algorithm has significantly improved the detection rate.

1. Identify the population and sample

- **Population:** All malicious activities/attacks that could be detected by the new cybersecurity system.
- **Sample:** 200 attacks tested using the new algorithm.

2. Identify the parameter of interest

- p = True detection rate of the new algorithm.

3. Calculate the point estimate

$$\hat{p} = \frac{200194}{200000} = 0.97$$

Therefore, the sample detection rate is **97%**.

4. State the null hypothesis H0

$$H_0: p = 0.95$$

The new algorithm has the same detection rate as the old system.

5. State the alternative hypothesis H1

$$H_1: p > 0.95$$

The new algorithm has improved the detection rate.

6. Select the significance level

$$\alpha = 0.05$$

7. Calculate the test statistic

For a proportion test:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)}} = \frac{0.97 - 0.95}{\sqrt{0.95(0.05)}} \approx 1.30$$

95% Confidence Interval:

$$0.9464 < p < 0.9936$$

or approximately **94.64% to 99.36%**.

8. Calculate the p-value

For the right-tailed test:

$$p\text{-value} \approx 0.0972$$

9. Compare p-value with α

$$0.0972 > 0.05$$

Therefore, **fail to reject H0**.

10. Industry-oriented conclusion

Although the new cybersecurity algorithm achieved a higher sample detection rate of **97% compared with 95%**, the improvement is **not statistically significant at the 5% significance level**. Therefore, based on this sample, the cybersecurity company does not have sufficient statistical evidence to conclude that the new algorithm significantly improves attack detection.

17. Healthcare – Hypothesis Testing Using p-value

Problem: The previous average patient waiting time was 42 minutes. After implementing a new scheduling system, a sample of 64 patients has an average waiting time of 38 minutes with a standard deviation of 12 minutes.

1. Identify the population and sample

- **Population:** All patients using the hospital's new appointment scheduling system.
- **Sample:** 64 randomly selected patients.

2. Identify the parameter of interest

- μ = True mean patient waiting time after implementing the new system.

3. Calculate the point estimate

$\bar{x}=38$ minutes

The estimated average waiting time is **38 minutes**.

4. State the null hypothesis H_0

$$H_0: \mu = 42$$

The new system has not reduced the average waiting time.

5. State the alternative hypothesis H_1

$$H_1: \mu < 42$$

The new system reduces the average waiting time.

6. Select the significance level

$$\alpha = 0.05$$

7. Calculate the test statistic

Since the population standard deviation is not given, use a one-sample t-test.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{40.38 - 42}{12/\sqrt{64}} = \frac{-1.62}{1.5} = -1.08$$

Degrees of freedom:

$$df = 64 - 1 = 63$$

95% Confidence Interval:

$$35.00 < \mu < 41.00$$

approximately **35.00 to 41.00 minutes.**

8. Calculate the p-value

For $t = -1.08$ with $df = 63$:

$$p\text{-value} \approx 0.1429$$

9. Compare p-value with α

$$0.1429 > 0.05$$

Therefore, **reject H_0 .**

10. Industry-oriented conclusion

The hospital has sufficient statistical evidence to conclude that the new appointment scheduling system **significantly reduces patient waiting time**. The average waiting time decreased from 42 minutes to 38 minutes, indicating that the new system improves hospital service efficiency.

18. Social Media Analytics – Hypothesis Testing Using p-value

Problem: A social media platform claims that the average daily usage time is 3 hours. A sample of 100 users has a mean usage time of 3.2 hours and standard deviation of 0.8 hours.

1. Identify the population and sample

- **Population:** All users of the social media platform.
- **Sample:** 100 randomly selected users.

2. Identify the parameter of interest

- μ = True mean daily usage time of all users.

3. Calculate the point estimate

$\bar{x}=3.2$ hours

The estimated average daily usage is **3.2 hours**.

4. State the null hypothesis H_0

$H_0:\mu=3$

The company's claim is correct.

5. State the alternative hypothesis H_1

$H_1:\mu \neq 3$

The true average usage time is different from 3 hours.

6. Select the significance level

$$\alpha=0.05$$

7. Calculate the test statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{3.2 - 3}{0.8/\sqrt{100}} = 2.50$$

Degrees of freedom:

$$df = 100 - 1 = 99$$

95% Confidence Interval:

$$3.041 < \mu < 3.359$$

approximately **3.04 to 3.36 hours**.

8. Calculate the p-value

For a two-tailed test with $t=2.50$ and $df=99$:

$$p\text{-value} \approx 0.014$$

9. Compare p-value with α

$$0.014 < 0.05$$

Therefore, **reject H_0** .

10. Industry-oriented conclusion

There is sufficient statistical evidence to reject the social media company's claim that the average daily usage time is exactly 3 hours. The sample indicates that users spend significantly different—and in this case higher—time on the platform, with an average of **3.2 hours per day**.

19. Machine Learning – Hypothesis Testing Using p-value

Problem: The existing model has an accuracy of 82%. A new model achieves 86% accuracy on a test sample of 500 observations. Test whether the improvement is statistically significant.

1. Identify the population and sample

- **Population:** All future observations that may be classified by the new machine learning model.
- **Sample:** 500 randomly selected test observations.

2. Identify the parameter of interest

- p = True classification accuracy of the new model.

3. Calculate the point estimate

$$\hat{p}=0.86$$

Therefore, the sample accuracy is **86%**.

4. State the null hypothesis H_0

$$H_0:p=0.82$$

The new model has the same accuracy as the existing model.

5. State the alternative hypothesis H_1

$$H_1:p>0.82$$

The new model has better accuracy than the existing model.

6. Select the significance level

$$\alpha=0.05$$

7. Calculate the test statistic

For a proportion test:

$$Z=\frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)}}=\frac{0.86-0.82}{\sqrt{0.82(1-0.82)}}\approx 2.33$$

95% Confidence Interval for the new model accuracy:

$$0.8296 < p < 0.8904$$

or approximately **82.96% to 89.04%**.

8. Calculate the p-value

For the right-tailed test:

$$p\text{-value} \approx 0.010$$

9. Compare p-value with α

$$0.010 < 0.05$$

Therefore, **reject H_0** .

10. Industry-oriented conclusion

The new machine learning model's accuracy of **86%** is a **statistically significant improvement over the existing 82% accuracy** at the 5% significance level. Therefore, the data science team has sufficient evidence to support deploying or further evaluating the new classification model.

20. Manufacturing / Quality Control – Confidence Interval + p-value

Problem: A manufacturer claims that the defect rate is 2%. From 2,000 randomly selected products, 55 are defective. Test the claim using both confidence interval and p-value approaches.

1. Identify the population and sample

- **Population:** All products produced by the manufacturer.
- **Sample:** 2,000 randomly selected products.

2. Identify the parameter of interest

- p = True proportion of defective products.

3. Calculate the point estimate

$$\hat{p} = \frac{200055}{72000} = 0.0275$$

Therefore, the sample defect rate is:

2.75%

4. State the null hypothesis H_0

$$H_0: p = 0.02$$

The true defect rate is 2%.

5. State the alternative hypothesis H_1

$$H_1: p \neq 0.02$$

The true defect rate is different from 2%.

6. Select the significance level

$$\alpha = 0.05$$

For the confidence interval approach, use a **95% confidence interval**.

7. Calculate the confidence interval and test statistic

(a) 95% Confidence Interval Approach

$$CI = \hat{p} \pm 1.96 \sqrt{\hat{p}(1-\hat{p})} \quad CI = 0.0275 \pm 1.96 \sqrt{0.0275(0.9725)}$$

$$CI = (0.0203, 0.0347)$$

Therefore, the **95% confidence interval** is approximately:

2.03% to 3.47%

Since **2% is not within the confidence interval**, reject the manufacturer's claim.

(b) Test Statistic

$$Z = \frac{n\hat{p} - np_0}{\sqrt{np_0(1-p_0)}} = \frac{20000(0.0275) - 20000(0.02)}{\sqrt{20000(0.02)(0.98)}} \approx 2.40$$

8. Calculate the p-value

Since this is a two-tailed test:

$$p\text{-value} \approx 0.0166$$

9. Compare p-value with α

$$0.0166 < 0.05$$

Therefore, **reject H_0** .

10. Industry-oriented conclusion

Both the **95% confidence interval approach** and the **p-value approach** give the same conclusion. There is sufficient statistical evidence to reject the manufacturer's claim that the defect rate is only 2%.

The observed defect rate is **2.75%**, and the confidence interval suggests that the true defect rate is likely between approximately **2.03% and 3.47%**. Therefore, the manufacturing company should investigate and improve its quality-control process because the actual defect rate appears to be significantly different from—and higher than—the claimed 2%.